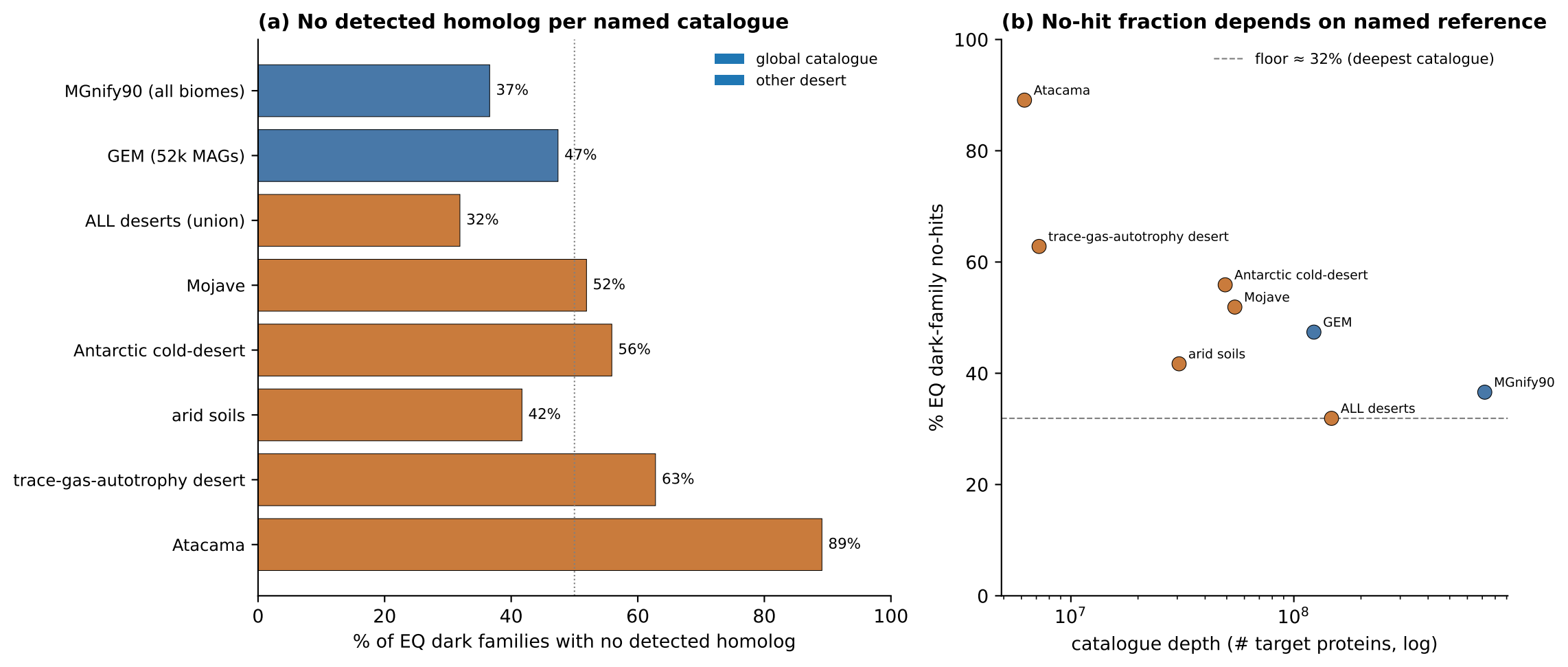
(a) No detected homolog per named catalogue
global catalogue
other desert
MGnify90 (all biomes)
GEM (52k MAGs)
ALL deserts (union)
Mojave
Antarctic cold-desert
arid soils
trace-gas-autotrophy desert
Atacama
37%
47%
32%
52%
56%
42%
63%
89%
0
20
40
60
80
100
% of EQ dark families with no detected homolog
(b) No-hit fraction depends on named reference
floor ≈ 32% (deepest catalogue)
% EQ dark-family no-hits
catalogue depth (# target proteins, log)
10^7
10^8
Atacama
trace-gas-autotrophy desert
Antarctic cold-desert
Mojave
GEM
arid soils
MGnify90
ALL deserts